Iman Rahmaty

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Research Areas: Mobile Edge Computing (MEC), Edge Intelligence, Deep Reinforcement Learning (DRL), Multi-Agent Reinforcement Learning (MARL), Collaborative Computation Offloading, Resource Allocation and Optimization, Distributed Decision-Making under Partial Observability, Intelligent Networked Systems.

EDUCATION

Master of Science in Computer Engineering/Computer Networks Graduated Sep. 2022

GPA: 87/100 Sharif University of Technology (SUT), Tehran, Nationally ranked no. 1 in Iran

Thesis: A Distributed Task Offloading Algorithm for Resource-Constrained MEC Networks Using DRL

Supervisor: Prof. Ali Movaghar

Bachelor of Science in Industrial Engineering Graduated Sep. 2019

Khajeh Nasir Toosi University of Technology (KNTU), Tehran, Nationally ranked no. 5 in Iran

RESEARCH EXPERIENCE

Research Engineer at EdgeAI Laboratory 2022 - Present

Supervisor: Prof. Hamed Shah-Mansouri Department of Electrical Engineering, SUT

- Investigate multi-agent DRL (MADRL) for collaborative computation offloading in MEC, focusing on partial observability, agent heterogeneity, and non-stationary system dynamics.
- Model interdependent offloading decisions as a decentralized partially observable Markov decision process (Dec-POMDP) and explore centralized training with decentralized execution (CTDE).
- Develop and evaluate coordinated learning policies to balance workloads and exploit computing collaboration among end devices and edge nodes.

Research Assistant at Performance and Dependability Laboratory (PDL) 2019-2022

Supervisor: Prof. Ali Movaghar Department of Computer Engineering, SUT

- Formulated distributed offloading in resource-constrained MEC as a Markov decision process (MDP), accounting for dynamic task arrivals, computational demands, deadlines, and workload fluctuations.
- Developed a DRL-based computation offloading algorithm based on Deep Q-Networks (DQN), incorporating Double Q-learning, a dueling network architecture, and LSTM-based workload history to learn independent offloading policies, capture time-varying edge-node loads, and optimize long-term QoE.
- Designed and implemented simulation experiments to evaluate QECO against existing approaches, achieving up to 37% higher QoE, with faster convergence than DQN and Double DQN.

TEACHING AND ACADEMIC SERVICE

Reviewer, IEEE Internet of Things Journal IEEE, 2025, 2026

Reviewer, 27th International Computer Conference CSICC, 2022

Teaching Assistants

- Performance Evaluation of Computer Systems (Head TA) SUT, 2020-2022
- Software Defined Networking (Head TA) SUT, 2022
- Verification of Reactive Systems SUT, 2021
- Wireless Networking, Theory of Machines and Languages SUT, 2020

PUBLICATION

- **I. Rahmaty**, H. Shah-Mansouri, A. Movaghar, "QECO: A QoE-Oriented Computation Offloading Algorithm based on Deep Reinforcement Learning for Mobile Edge Computing", *IEEE Transactions on Network Science and Engineering*, vol. 12, no. 4, pp. 3118-3130, July-Aug. 2025.
- **I. Rahmaty**, H. Shah-Mansouri, M. Hosseini, A. Movaghar, "A Collaborative Computation Offloading Algorithm based on Multi-Agent Deep Reinforcement Learning for Partially Observable Mobile Edge Computing", *Work in progress.*

SELECTED PROJECTS

- **Multi-Agent Actor-Critic for Interdependent Decision-Making** EdgeAI, 2025
  Developing a MARL algorithm for interdependent offloading decisions in non-stationary MEC, formulating agent interactions as a Dec-POMDP and investigating CTDE to learn coordinated strategies.
- **Cloud–Edge–End Collaborative Computing Simulation** EdgeAI, 2024
  Implemented a time-driven MEC simulator supporting device-to-device and edge-to-edge communication to evaluate collaborative strategies, interdependent decisions, and system-wide resource utilization.
- **Dueling Double DQN (D3QN)-Based Distributed Decision-Making** EdgeAI, 2023
  Developed a D3QN-based offloading algorithm for dynamic MEC, formulating independent decision-making as an MDP to maximize personalized user QoE. Integrated Double DQN, dueling architecture, experience replay, and target networks to improve value estimation and learning stability.
- **LSTM-Based Edge-Server Workload Estimation** PDL, 2022
  Developed an LSTM model to capture temporal workload patterns, informing MEC offloading decisions.
- **Resource-Constrained MEC Simulation** PDL, 2021
  Built a time-driven simulator of queueing MEC under stochastic conditions and resource constraints.
- **M/M/1/K Queueing System Simulation** SUT, 2020
  Implemented an event-driven simulator for FCFS, PS, and discriminatory PS queueing disciplines.

SKILLS

- **Professional:** Stochastic Systems Modeling, Simulation, and Performance Evaluation; DRL Algorithm Design and Implementation; Resource Allocation Optimization; Experimental Design, Benchmarking
- **Programming & ML:** Python, C++, R, SQL, Bash; PyTorch, TensorFlow, Keras, Scikit-learn
- **Simulation & Tools:** Ns-3, Mininet, Linux, Git, NumPy, Pandas, Matplotlib, Jupyter, Flask, LaTeX
- **Languages:** Persian (Native), English (TOEFL iBT: 108/120, Jan 2024)

HONORS

❖ Ranked top 10% of M.Sc. students in the Dept. of Computer Engineering at SUT (Class of 2019) 2022
❖ Ranked 55th among 60,000 in the nationwide M.Sc. entrance exam in Computer Engineering. 2019
❖ Ranked top 1% among 180,000 in the nationwide B.Sc. entrance exam in Mathematics and Physics. 2014
❖ Achieving the 3rd position in the RoboCup Competition (IranOpen) 2012

SELECTED CERTIFICATIONS

Interactive Learning (RL and Bandit Algorithms) Tehran Institute for Advanced Studies, 2021
Machine Learning, Deep Learning, and Data Science with Python Start-Tech Academy, 2020
Advanced Python Topics Remis Arjang Institute, 2018
Linux Professional Institute (LPIC-1) Anisa Iran Linux House, 2017